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Hyper-Panconnectedness of the Locally Twisted Cube *

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Abstract

A graph G is said to be panconnected (Alavi, 1975) if for any two distinct nodes x and y, it has a path of length l joining x and y for any integer l satisfying $d_G(x,y) \leq l \leq |V(G)| - 1$, where $d_G(x,y)$ denotes the distance between nodes x and y in G, and $|V(G)|$ is the total number of nodes of G. In this paper, we first introduce an extension of the panconnectedness property mentioned above: A graph G is said to be hyper-panconnected if for any two distinct nodes x and y of G, it contains a Hamiltonian path P such that $P(1) = x$ and $P(l+1) = y$ for any integer l satisfying $d_G(x,y) \leq l \leq |V(G)| - 1$, where $P(i)$ represents the ith node sequentially traversed by path P. As the importance of panconnectedness property for data communication between units in multiprocessor interconnected systems, the hyper-panconnectedness admits more flexible message broadcasting in network communication. Then we study this property with respect to the class of locally twisted cubes, which is a popular variant of the hypercube architecture.

1 Introduction

A massive multiprocessor system connects a large number of processors on the basis of an interconnection network so that the interconnection network comes into a important factor affecting system performance [12, 15, 21]. In terms of network analysis, the topological structure of an interconnection network can be modeled as a graph whose nodes and links represent processors and communication buses, respectively. Among many kinds of network topologies, the binary n-cube [20] (also called hypercube) is one of the most popular networks for parallel and distributed computing. Not only is it ideally suited to both special-purpose and general-purpose tasks, but it can also efficiently simulate many other networks [12, 15, 21]. However, the hypercube architecture is bipartite, so it cannot make the best use of hardware resources in the design of parallel algorithms. One severe disadvantage of the hypercube is that it has the largest diameter among cube family. To compensate for this drawback, many researchers [1, 6, 7, 23] have tried to fashion the hypercube into ones with lower diameters. One such network topology is the locally twisted cube, which was first proposed by Yang et al. [23]. An n-dimensional locally twisted cube is derived from the binary n-cube by changing some connections of links. Its diameter is only $\lceil \frac{n+3}{2} \rceil$ if $n \geq 5$, about half of the hypercube's. Besides, the locally twisted cube has many attractive properties. For example, it can embed paths of odd and even lengths [16] and many-to-many disjoint path covers [18]; it has more cycles than the hypercube [22, 17]. The definition of locally twisted cubes will be presented in the next section.

Throughout this paper graphs are finite, simple, and undirected. Some important graph-theoretic definitions and notations will be introduced in advance. For those not defined here, however, we follow the standard terminology given by Bondy and Murty [4]. An *undirected graph* G is an ordered pair (V,E), where V is a nonempty set, and E is a subset of $\{\{u,v\} | \{u,v\}$ is a 2-element subset of $V\}$. The

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set V is called the *node set* of G, and the set E is called the *link set* of G. For convenience, we denote the node set and the link set of G by $V(G)$ and $E(G)$, respectively.

Two nodes u and v of G are *adjacent* if $\{u,v\} \in E(G)$. The *degree* of a node u in G is the number of links incident to u. A graph G is *k-regular* if all its nodes have the same degree k. For any node v of G, its *neighborhood* $N_G(v)$ is defined by $N_G(v) = \{u \in V(G) \mid \{v,u\} \in E(G)\}$. A graph H is a *subgraph* of a graph G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. Let S be a nonempty subset of $V(G)$. The subgraph of G *induced* by S is a subgraph of G with node set S, whose link set consists of all the links joining any two nodes in S.

A *path* P of length $k \geq 1$ from node x to node y in a graph G is a sequence of distinct nodes $\langle v_1, v_2, \ldots, v_{k+1}\rangle$ such that $v_1 = x$, $v_{k+1} = y$, and $\{v_i, v_{i+1}\} \in E(G)$ for every $1 \leq i \leq k$. Moreover, a path of length 0, consisting of a single node x, is denoted by $\langle x\rangle$. For convenience, we write P as $\langle v_1, v_2, \ldots, v_i, Q, v_j, \ldots, v_{k+1}\rangle$, where $i \leq j$ and $Q = \langle v_i, \ldots, v_j\rangle$. The ith node of P is denoted by $P(i)$; i.e., $P(i) = v_i$. In particular, let $rev(P)$ represent the reverse of P; that is, $rev(P) = \langle v_{k+1}, v_k, \ldots, v_1\rangle$. We use $\ell(P)$ to denote the *length* of path P. The *distance* between two distinct nodes u and v of G, denoted by $d_G(u,v)$, is the length of the shortest path between u and v in G. A *cycle* is a path with at least three nodes such that the last node is adjacent to the first one. For clarity, a cycle of length $k \geq 3$ is represented by $\langle v_1, v_2, \ldots, v_k, v_1\rangle$. A path (or cycle) in a graph G is a *Hamiltonian path* (or *Hamiltonian cycle*) of G if it spans G. A graph G is *Hamiltonian* if it has a Hamiltonian cycle; a graph G is *Hamiltonian connected* if it contains a Hamiltonian path to join any pair of distinct nodes.

In recent years, many research results about cycle embedding have been focused on exploring the properties of pancyclicity [5, 8, 13, 14, 16, 19, 22]. A graph G is called *pancyclic* [3] if it contains a cycle of length l for each integer l from 3 to $|V(G)|$ inclusive. More specifically, a graph G is called *link-pancyclic* (respectively, *node-pancyclic*) if its any link (respectively, node) lies on a cycle of length l for every $3 \leq l \leq |V(G)|$. On the other hand, a graph G is said to be *panconnected* [2] if for any two distinct nodes x and y, it has a path of length l to join x and y for any integer l satisfying $d_G(x,y) \leq l \leq |V(G)| - 1$. It is easy to see that every panconnected graph must be pancyclic, link-pancyclic, and node-pancyclic. In this paper, we introduce an extension of panconnectedness property: A graph G is said to be *hyper-panconnected* if for any two distinct nodes x and y of G, it contains a Hamiltonian path P such that $P(1) = x$ and $P(l+1) = y$ for any integer l satisfying $d_G(x,y) \leq l \leq |V(G)| - 1$, As the significance of panconnectedness property for data communication between units in multiprocessor interconnected systems, the hyper-panconnectedness admits more flexible message broadcasting in network communication. Then we study this property with respect to the class of locally twisted cubes [23], which is a popular variant of the hypercube architecture [20].

The rest of this paper is organized as follows. In Section 2, the definition of locally twisted cubes is introduced. In Section 3, the main theorem and its proof are given. Finally, some concluding remarks are described in Section 4.

2 The locally twisted cube and its properties

The n-dimensional locally twisted cube, denoted by LTQ_n, has 2^n nodes, each of which corresponds to an n-bit binary string. Its definition is given as below.

Definition 1. *[23] The n-dimensional locally twisted cube LTQ_n is recursively constructed as follows:*

1. *$LTQ_1 = (\{0,1\}, \{\{0,1\}\})$.*
2. *$LTQ_2 = (\ \{00, 01, 10, 11\}, \{\ \{\ 00,\ 01\ \}, \{\ 00,\ 10\ \}, \{\ 01,\ 11\ \}, \{\ 10,\ 11\ \}\ \}\ )$.*
3. *For $n \geq 3$, let LTQ^0_{n-1} and LTQ^1_{n-1} be two copies of LTQ_{n-1} with $V(LTQ^0_{n-1}) = \{\ 0\ x_{n-1}\ x_{n-2}\ \ldots\ x_1 \mid x_i = 0$ or 1 for $1 \leq i \leq n-1\ \}$ and $V(LTQ^1_{n-1}) = \{1y_{n-1}y_{n-2}\ldots y_1 \mid y_i = 0$ or 1 for $1 \leq i \leq n-1\}$. Then LTQ_n is formed by connecting LTQ^0_{n-1} and LTQ^1_{n-1} with 2^{n-1} links so that a node $\mathbf{x} = 0x_{n-1}x_{n-2}\ldots x_1$ in LTQ^0_{n-1} is adjacent to a node $\mathbf{y} = 1y_{n-1}y_{n-2}\ldots y_1$ in LTQ^1_{n-1} if and only if*
   - *(a) $y_{n-1} = x_{n-1} \oplus x_1$, where $\oplus$ denotes the addition modulo 2, and*
   - *(b) $y_i = x_i$ for every $1 \leq i \leq n-2$.*

The LTQ_2 is really formed by connecting two copies of LTQ_1. For the sake of convenience, we

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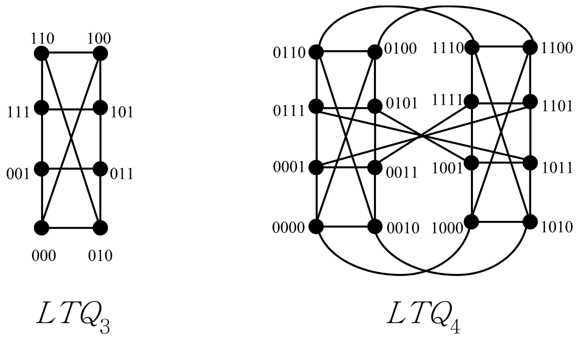

Figure 1: Illustration of LTQ_3 and LTQ_4.

denote this recursive construction by $LTQ_n = LTQ^0_{n-1} \otimes LTQ^1_{n-1}$ for $n \geq 2$. From the above definitions, LTQ_2 is nothing but a cycle of length 4. Furthermore, LTQ_3 is isomorphic to circulant graph $C(8; 4)$. We depict LTQ_3 and LTQ_4 in Figure 1. Yang et al. [23] proved that LTQ_n is n-connected and has diameter $\lceil \frac{n+3}{2} \rceil$ (respectively, $n-1$) for $n \geq 5$ (respectively, $n = 3, 4$). The locally twisted cube has also received many researchers' attention [10, 11, 16, 17, 22, 23].

Suppose that $\mathbf{u} = u_n u_{n-1} \dots u_1$ denotes any node in LTQ_n for $n \geq 3$. Then $\mathbf{u}$ is said to be adjacent to a node $\mathbf{v} = v_n v_{n-1} \dots v_1$ along the 1st dimension if $u_j = v_j$ for all $j \in \{2, 3, \cdots, n\}$; $\mathbf{u}$ is adjacent to $\mathbf{v}$ along the 2nd dimension if $u_j = v_j$ for all $j \in \{1, 3, \cdots, n\}$; $\mathbf{u}$ is adjacent to $\mathbf{v}$ along the ith dimension, $3 \leq i \leq n$, if the following three conditions are all satisfied: (i) $u_i \neq v_i$, (ii) $u_j = v_j$ for all $j \in \{1, \cdots, i-2, i+1, \cdots, n\}$, (iii) $u_{i-1} = v_{i-1} \oplus v_1$. Such a commutative relation between adjacent nodes $\mathbf{u}$ and $\mathbf{v}$ is denoted by $\mathbf{u} \overset{i}{\leftrightarrow} \mathbf{v}$. Moreover, $\mathbf{u}$ is said to be the i-neighbor of $\mathbf{v}$, and vice versa. For convenience, the i-neighbor of node $\mathbf{u}$ is denoted by $(\mathbf{u})^i$. The link $\{\mathbf{u}, (\mathbf{u})^i\}$ is called *i-dimensional*. It is easy to see that $\mathbf{v} = (\mathbf{u})^i$ if and only if $\mathbf{u} = (\mathbf{v})^i$. The next two lemmas show how to locate cycles of length 4 or 5 in LTQ_n, respectively.

Lemma 1. *Let $\{\mathbf{u}, \mathbf{v}\}$ be any n-dimensional link in LTQ_n for $n \geq 3$. Then the set of nodes $\{\mathbf{u}, \mathbf{v}, (\mathbf{u})^i, (\mathbf{v})^i\}$ induces a cycle of length 4 if $2 \leq i \leq n-1$.*

Lemma 2. *Let $\mathbf{u} = u_n u_{n-1} \cdots u_1$ be any node in LTQ_n for $n \geq 3$.*

1. *The set of nodes $\{ \mathbf{u}, (\mathbf{u})^1, (\mathbf{u})^n, ((\mathbf{u})^1)^n, ((\mathbf{u})^n)^{n-1} \}$ induces a cycle of length 5 if $u_1 = 0$.*
2. *The set of nodes $\{ \mathbf{u}, (\mathbf{u})^1, (\mathbf{u})^n, ((\mathbf{u})^1)^n, ((\mathbf{u})^n)^1 \}$ induces a cycle of length 5 if $u_1 = 1$.*

In [23], Yang et al. proposed a shortest path routing algorithm $\mathbf{ROUTE}(Msg, \mathbf{u}, \mathbf{v})$ for LTQ_n, which implies the following two lemmas.

Lemma 3. *For $n \geq 2$, let $\mathbf{u}$ and $\mathbf{v}$ be any two different nodes of LTQ_n such that $\{\mathbf{u}, \mathbf{v}\} \subset V(LTQ^i_{n-1})$, $i \in \{0, 1\}$. Then $d_{LTQ_n}(\mathbf{u}, \mathbf{v}) = d_{LTQ^i_{n-1}}(\mathbf{u}, \mathbf{v})$.*

Lemma 4. *For $n \geq 2$, let $\mathbf{u}$ and $\mathbf{v}$ denote any two nodes in LTQ_n such that $\mathbf{u}$ is in LTQ^0_{n-1} and $\mathbf{v}$ is in LTQ^1_{n-1}. Then $d_{LTQ_n}(\mathbf{u}, (\mathbf{v})^n) = d_{LTQ_n}(\mathbf{u}, \mathbf{v}) - 1$ or $d_{LTQ_n}((\mathbf{u})^n, \mathbf{v}) = d_{LTQ_n}(\mathbf{u}, \mathbf{v}) - 1$.*

A Hamiltonian graph G is said to be *f-fault-tolerant Hamiltonian* if $G - F$ remains Hamiltonian for every $F \subseteq V(G) \cup E(G)$ with $|F| \leq f$. A Hamiltonian connected graph G is said to be *f-fault-tolerant Hamiltonian connected* if $G - F$ remains Hamiltonian connected for every $F \subseteq V(G) \cup E(G)$ with $|F| \leq f$. The results in [9, 17, 19] imply the following lemma.

Lemma 5. *For any integer n, $n \geq 3$, LTQ_n is $(n-2)$-fault-tolerant Hamiltonian and $(n-3)$-fault-tolerant Hamiltonian connected.*

3 Hyper-Panconnectedness of LTQ_n

Theorem 1. *Let $\mathbf{x}$ and $\mathbf{y}$ be any two nodes of LTQ_n, $n \geq 4$, and let l be any integer with $d_{LTQ_n}(\mathbf{x}, \mathbf{y}) + 2 \leq l \leq 2^n - 1$. There exists a Hamiltonian path P in LTQ_n such that $P(1) = \mathbf{x}$ and $P(l+1) = \mathbf{y}$.*

Proof. This theorem is proved by induction on n. Firstly, the correctness of the induction base on LTQ_4 can be verified by brute force with a computer program. The inductive hypothesis is that the statement holds for any LTQ_k, $4 \leq k \leq n-1$. Then we need to show that LTQ_n has a Hamiltonian path P such that $P(1) = \mathbf{x}$ and $P(l+1) = \mathbf{y}$. Since $LTQ_n = LTQ^0_{n-1} \otimes LTQ^1_{n-1}$, we assume, without loss of generality, that $\mathbf{x}$ is in LTQ^0_{n-1}. The following three cases are distinguished.

Case 1: Suppose that $\mathbf{y}$ is in LTQ^0_{n-1}. By Lemma 3, we have $d_{LTQ_n}(\mathbf{x}, \mathbf{y}) = d_{LTQ^0_{n-1}}(\mathbf{x}, \mathbf{y})$. Thus, the following three subcases have to be considered.

Subcase 1.1: Suppose that $d_{LTQ_n}(\mathbf{x}, \mathbf{y}) + 2 \leq l \leq 2^{n-1} - 1$. By the inductive hypothesis, there

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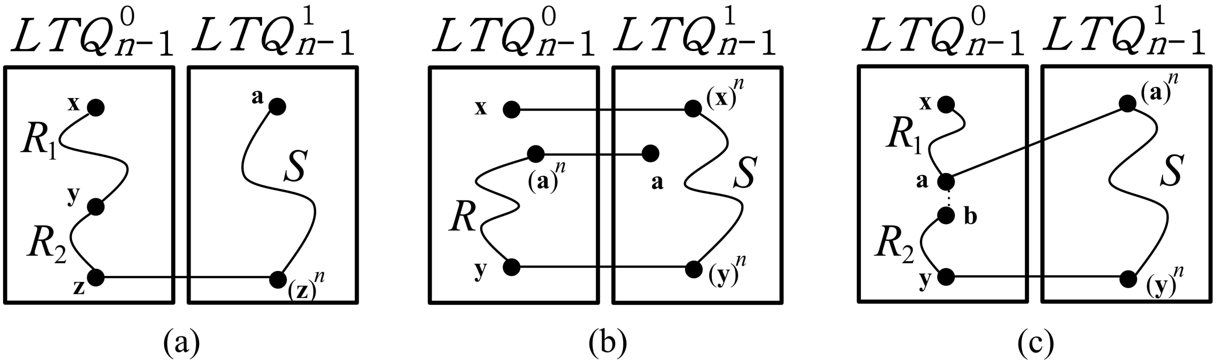

Figure 2: Case 1 in the proof for Theorem 1. (A dashed line or a straight line represents a link.)

exists a Hamiltonian path R of LTQ^0_{n-1} such that $R(1) = \mathbf{x}$ and $R(l+1) = \mathbf{y}$. For convenience, path R can be written as $\langle \mathbf{x}, R_1, \mathbf{y}, R_2, \mathbf{z}\rangle$, where $\mathbf{z}$ is some node in LTQ^0_{n-1}. It is noticed that $\mathbf{z} = \mathbf{y}$ if $l = 2^{n-1} - 1$. Similarly, we can also find a Hamiltonian path S of LTQ^1_{n-1} joining $(\mathbf{z})^n$ to some node $\mathbf{a}$. Then $P = \langle \mathbf{x}, R_1, \mathbf{y}, R_2, \mathbf{z}, (\mathbf{z})^n, S, \mathbf{a}\rangle$ is a Hamiltonian path of LTQ_n with $P(1) = \mathbf{x}$ and $P(l+1) = \mathbf{y}$. See Figure 2(a) for illustration.

Subcase 1.2: Suppose that $l = 2^{n-1}$. Let $\mathbf{a}$ be any node in LTQ^1_{n-1} other than $(\mathbf{x})^n$ and $(\mathbf{y})^n$. By Lemma 5, there exists a Hamiltonian path S of $LTQ^1_{n-1} - \{\mathbf{a}\}$ joining $(\mathbf{x})^n$ to $(\mathbf{y})^n$. Similarly, there exists a Hamiltonian path R of $LTQ^0_{n-1} - \{\mathbf{x}\}$ joining $(\mathbf{a})^n$ to $\mathbf{y}$. Then $P = \langle \mathbf{x}, (\mathbf{x})^n, S, (\mathbf{y})^n, \mathbf{y}, R, (\mathbf{a})^n, \mathbf{a}\rangle$ is a Hamiltonian path of LTQ_n such that $P(1) = \mathbf{x}$ and $P(2^{n-1}+1) = \mathbf{y}$. Figure 2(b) illustrates this subcase.

Subcase 1.3: Suppose that $2^{n-1} + 1 \leq l \leq 2^n - 1$. By Lemma 5, there exists a Hamiltonian path R in LTQ^0_{n-1} joining $\mathbf{x}$ to $\mathbf{y}$. For clarity, the path R can be written as $\langle \mathbf{x}, R_1, \mathbf{a}, \mathbf{b}, R_2, \mathbf{y}\rangle$, where $\mathbf{a}$ and $\mathbf{b}$ are adjacent nodes satisfying $\ell(R_1) = l - 2^{n-1} - 1$. It is noticed that $\mathbf{x} = \mathbf{a}$ if $l = 2^{n-1} + 1$, and $\mathbf{b} = \mathbf{y}$ if $l = 2^n - 1$. Again, Lemma 5 ensures that LTQ^1_{n-1} has a Hamiltonian path S joining $(\mathbf{a})^n$ to $(\mathbf{y})^n$. Then $P = \langle \mathbf{x}, R_1, \mathbf{a}, (\mathbf{a})^n, S, (\mathbf{y})^n, \mathbf{y}, rev(R_2), \mathbf{b}\rangle$ is a Hamiltonian path of LTQ_n such that $P(1) = \mathbf{x}$ and $P(l+1) = \mathbf{y}$. See Figure 2(c) for illustration.

Case 2: Suppose that $\mathbf{y}$ is in LTQ^1_{n-1} and not adjacent to $\mathbf{x}$. By Lemma 3 and Lemma 4, we have $d_{LTQ^0_{n-1}}(\mathbf{x}, (\mathbf{y})^n) = d_{LTQ_n}(\mathbf{x}, \mathbf{y}) - 1$ or $d_{LTQ^1_{n-1}}((\mathbf{x})^n, \mathbf{y}) = d_{LTQ_n}(\mathbf{x}, \mathbf{y}) - 1$. The following subcases are distinguished.

Subcase 2.1: Suppose that $d_{LTQ_n}(\mathbf{x}, \mathbf{y}) + 2 \leq l \leq 2^{n-1} - 1$. Firstly, we assume that $d_{LTQ^0_{n-1}}(\mathbf{x}, (\mathbf{y})^n) = d_{LTQ_n}(\mathbf{x}, \mathbf{y}) - 1$. By the inductive hypothesis, there exists a Hamiltonian path R of LTQ^0_{n-1} with $R(1) = \mathbf{x}$ and $R(l) = (\mathbf{y})^n$. For clarity, the path R can be written as $\langle \mathbf{x}, R_1, (\mathbf{y})^n, \mathbf{a}, R_2, \mathbf{z}\rangle$, where $\mathbf{a}$ is a node adjacent to $(\mathbf{y})^n$, and $\mathbf{z}$ is some node in LTQ^0_{n-1}. It is noticed that $\mathbf{a} = \mathbf{z}$ if $l = 2^{n-1} - 1$. By Lemma 5, there exists a Hamiltonian path S of LTQ^1_{n-1} joining $\mathbf{y}$ to $(\mathbf{z})^n$. Then $P = \langle \mathbf{x}, R_1, (\mathbf{y})^n, \mathbf{y}, S, (\mathbf{z})^n, \mathbf{z}, rev(R_2), \mathbf{a}\rangle$ is a Hamiltonian path of LTQ_n such that $P(1) = \mathbf{x}$ and $P(l+1) = \mathbf{y}$. This subcase is illustrated in Figure 3(a).

Next, we assume that $d_{LTQ^1_{n-1}}((\mathbf{x})^n, \mathbf{y}) = d_{LTQ_n}(\mathbf{x}, \mathbf{y}) - 1$. By the inductive hypothesis, there exists a Hamiltonian path S of LTQ^1_{n-1} with $S(1) = (\mathbf{x})^n$ and $S(l) = \mathbf{y}$. The path S is written as $\langle (\mathbf{x})^n, S_1, \mathbf{y}, S_2, \mathbf{z}\rangle$, where $\mathbf{z}$ is some node in LTQ^1_{n-1}. By Lemma 5, there exists a Hamiltonian path R of $LTQ^0_{n-1} - \{\mathbf{x}\}$ joining $(\mathbf{z})^n$ to any node $\mathbf{a}$. Then, $P = \langle \mathbf{x}, (\mathbf{x})^n, S_1, \mathbf{y}, S_2, \mathbf{z}, (\mathbf{z})^n, R, \mathbf{a}\rangle$ is a Hamiltonian path of LTQ_n such that $P(1) = \mathbf{x}$ and $P(l+1) = \mathbf{y}$. See Figure 3(b) for illustration.

Subcase 2.2: Suppose that $2^{n-1} \leq l \leq 2^n - 2$. Let $h = l - 2^{n-1}$. By Lemma 5, LTQ^0_{n-1} has a Hamiltonian path R between $\mathbf{x}$ and $(\mathbf{y})^n$. For convenience, path R is written as $\langle \mathbf{x}, R_1, \mathbf{a}, \mathbf{b}, R_2, (\mathbf{y})^n\rangle$, where $\mathbf{a}$ and $\mathbf{b}$ are adjacent nodes with $\ell(R_1) = h$. It is noticed that $\mathbf{a} = \mathbf{x}$ if $l = 2^{n-1}$ and $\mathbf{b} = (\mathbf{y})^n$ if $l = 2^n - 2$. Obviously, LTQ^1_{n-1} has a Hamiltonian path S joining $(\mathbf{a})^n$ to $\mathbf{y}$. Then $P = \langle \mathbf{x}, R_1, \mathbf{a}, (\mathbf{a})^n, S, \mathbf{y}, (\mathbf{y})^n, rev(R_2), \mathbf{b}\rangle$ is a Hamiltonian path in LTQ_n with $P(1) = \mathbf{x}$ and $P(l+1) = \mathbf{y}$. See Figure 3(c) for illustration.

Subcase 2.3: Suppose that $l = 2^n - 1$. By Lemma 5, LTQ_n is Hamiltonian connected. Thus, there exists a Hamiltonian path P of LTQ_n joining

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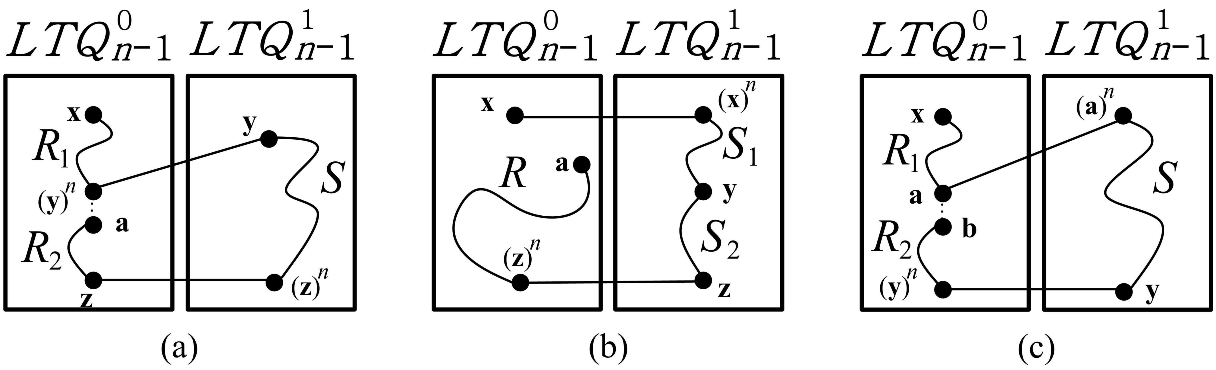

Figure 3: Case 2 in the proof for Theorem 1. (A dashed line or a straight line represents a link.)

$\mathbf{x}$ to $\mathbf{y}$.

Case 3: Suppose that $\mathbf{y}$ is in LTQ^1_{n-1} and adjacent to $\mathbf{x}$. Because $\mathbf{x}$ and $\mathbf{y}$ are adjacent, we have $3 \le l \le 2^n - 1$. The following subcases are distinguished.

Subcase 3.1: Suppose that $l = 3$. By Lemma 1, the set of nodes $\{\mathbf{x}, \mathbf{y}, (\mathbf{x})^2, (\mathbf{y})^2\}$ induces a cycle of length 4. By Lemma 5, there exists a Hamiltonian cycle H in $LTQ^0_{n-1} - \{\mathbf{x}, (\mathbf{x})^2\}$. We can write H as $\langle \mathbf{b}, R, \mathbf{z}, \mathbf{b} \rangle$, where $\mathbf{b}$ and $\mathbf{z}$ are adjacent nodes in $LTQ^0_{n-1} - \{\mathbf{x}, (\mathbf{x})^2\}$. By Lemma 5, there exists a Hamiltonian path S of $LTQ^1_{n-1} - \{(\mathbf{y})^2\}$ joining $\mathbf{y}$ to $(\mathbf{b})^n$. Then $P = \langle \mathbf{x}, (\mathbf{x})^2, (\mathbf{y})^2, \mathbf{y}, S, (\mathbf{b})^n, \mathbf{b}, R, \mathbf{z} \rangle$ is a Hamiltonian path in LTQ_n with $P(1) = \mathbf{x}$ and $P(4) = \mathbf{y}$. See Figure 4(a) for illustration.

Subcase 3.2: Suppose that $l = 4$. By Lemma 2, there exists a node $\mathbf{w}$ in LTQ^0_{n-1} such that the set of nodes $\{\mathbf{x}, \mathbf{y}, \mathbf{w}, (\mathbf{y})^1, ((\mathbf{y})^1)^n\}$ induces a cycle of length 5. By Lemma 5, there exists a Hamiltonian cycle H in $LTQ^0_{n-1} - \{\mathbf{x}, \mathbf{w}\}$. For clarity, H is written as $\langle ((\mathbf{y})^1)^n, \mathbf{v}, R, \mathbf{u}, ((\mathbf{y})^1)^n \rangle$, where $\mathbf{u}$ and $\mathbf{v}$ are some nodes adjacent to $((\mathbf{y})^1)^n$. By Lemma 5, there exists a Hamiltonian path S in $LTQ^1_{n-1} - \{(\mathbf{y})^1\}$ joining $\mathbf{y}$ to $(\mathbf{v})^n$. Then $P = \langle \mathbf{x}, \mathbf{w}, ((\mathbf{y})^1)^n, (\mathbf{y})^1, \mathbf{y}, S, (\mathbf{v})^n, \mathbf{v}, R, \mathbf{u} \rangle$ is a Hamiltonian path in LTQ_n with $P(1) = \mathbf{x}$ and $P(5) = \mathbf{y}$. See Figure 4(b) for illustration.

Subcase 3.3: Suppose that $5 \le l \le 2^{n-1}$. Let $h = l - 2$. Therefore, we have $3 \le h \le 2^{n-1} - 2$. Clearly, the set of nodes $\{\mathbf{x}, \mathbf{y}, (\mathbf{x})^2, (\mathbf{y})^2\}$ induces a cycle of length 4. By the inductive hypothesis, there exists a Hamiltonian path R in LTQ^0_{n-1} such that $R(1) = \mathbf{x}$ and $R(h+1) = (\mathbf{x})^2$. For clarity, the path R can be written as $\langle \mathbf{x}, R_1, (\mathbf{x})^2, \mathbf{v}, R_2, \mathbf{z} \rangle$, where $\mathbf{v}$ is some node adjacent to $(\mathbf{x})^2$, and $\mathbf{z}$ is some node in $V(LTQ^0_{n-1}) - \{\mathbf{x}, (\mathbf{x})^2\}$. It is noticed that $\mathbf{z} = \mathbf{v}$ if $h = 2^{n-1} - 2$. By Lemma 5, there exists a Hamiltonian path S of $LTQ^1_{n-1} - \{(\mathbf{y})^2\}$ joining $\mathbf{y}$ to $(\mathbf{z})^n$. Then $P = \langle \mathbf{x}, R_1, (\mathbf{x})^2, (\mathbf{y})^2, \mathbf{y}, S, (\mathbf{z})^n, \mathbf{z}, rev(R_2), \mathbf{v} \rangle$ is a Hamiltonian path in LTQ_n with $P(1) = \mathbf{x}$ and $P(l+1) = \mathbf{y}$. See Figure 4(c) for illustration.

Subcase 3.4: Suppose that $l = 2^{n-1} + 1$. Obviously, the set of nodes $\{\mathbf{x}, \mathbf{y}, (\mathbf{x})^2, (\mathbf{y})^2\}$ induces a cycle of length 4. By the inductive hypothesis, there exists a Hamiltonian path R in LTQ^0_{n-1} joining $\mathbf{x}$ and $(\mathbf{x})^2$. By Lemma 5, there exists a Hamiltonian path S of $LTQ^1_{n-1} - \{(\mathbf{y})^2\}$ joining $\mathbf{y}$ to some node $\mathbf{z}$. Then $P = \langle \mathbf{x}, R, (\mathbf{x})^2, (\mathbf{y})^2, \mathbf{y}, S, \mathbf{z} \rangle$ is a Hamiltonian path of LTQ_n with $P(1) = \mathbf{x}$ and $P(2^{n-1}+2) = \mathbf{y}$. Figure 4(d) illustrates this subcase.

Subcase 3.5: Suppose that $l = 2^{n-1} + 2$. Let $\langle \mathbf{y}, \mathbf{u}, \mathbf{v} \rangle$ represent any path of length 2 in LTQ^1_{n-1}. By Lemma 5, LTQ^0_{n-1} has a Hamiltonian path R joining $\mathbf{x}$ to $(\mathbf{v})^n$, and $LTQ^1_{n-1} - \{\mathbf{u}, \mathbf{v}\}$ has a Hamiltonian cycle H. For clarity, H is written as $\langle \mathbf{y}, S, \mathbf{z}, \mathbf{y} \rangle$, where $\mathbf{z}$ is some node adjacent to $\mathbf{y}$. Then $P = \langle \mathbf{x}, R, (\mathbf{v})^n, \mathbf{v}, \mathbf{u}, \mathbf{y}, S, \mathbf{z} \rangle$ is a Hamiltonian path of LTQ_n with $P(1) = \mathbf{x}$ and $P(2^{n-1}+3) = \mathbf{y}$. See Figure 4(e).

Subcase 3.6: Suppose that $2^{n-1} + 3 \le l \le 2^n - 1$. Let $h = l - 2^{n-1}$. Hence, we have $3 \le h \le 2^{n-1} - 1$. Obviously, the set of nodes $\{\mathbf{x}, \mathbf{y}, (\mathbf{x})^2, (\mathbf{y})^2\}$ induces a cycle of length 4. By Lemma 5, there exists a Hamiltonian path R in LTQ^0_{n-1} joining $\mathbf{x}$ and $(\mathbf{x})^2$. By the inductive hypothesis, there exists a Hamiltonian path S of LTQ^1_{n-1} with $S(1) = (\mathbf{y})^2$ and $S(h+1) = \mathbf{y}$. For convenience, S is written as $\langle (\mathbf{y})^2, S_1, \mathbf{y}, S_2, \mathbf{z} \rangle$, where $\mathbf{z}$ is some node of $V(LTQ^1_{n-1}) - \{\mathbf{y}, (\mathbf{y})^2\}$. It is noticed that $\mathbf{z} = \mathbf{y}$ if $h = 2^{n-1} - 1$. Then $P = \langle \mathbf{x}, R, (\mathbf{x})^2, (\mathbf{y})^2, S_1, \mathbf{y}, S_2, \mathbf{z} \rangle$ turns out to be a Hamiltonian path of LTQ_n with $P(1) = \mathbf{x}$ and $P(l+1) = \mathbf{y}$. See Figure 4(f) for illustration. □

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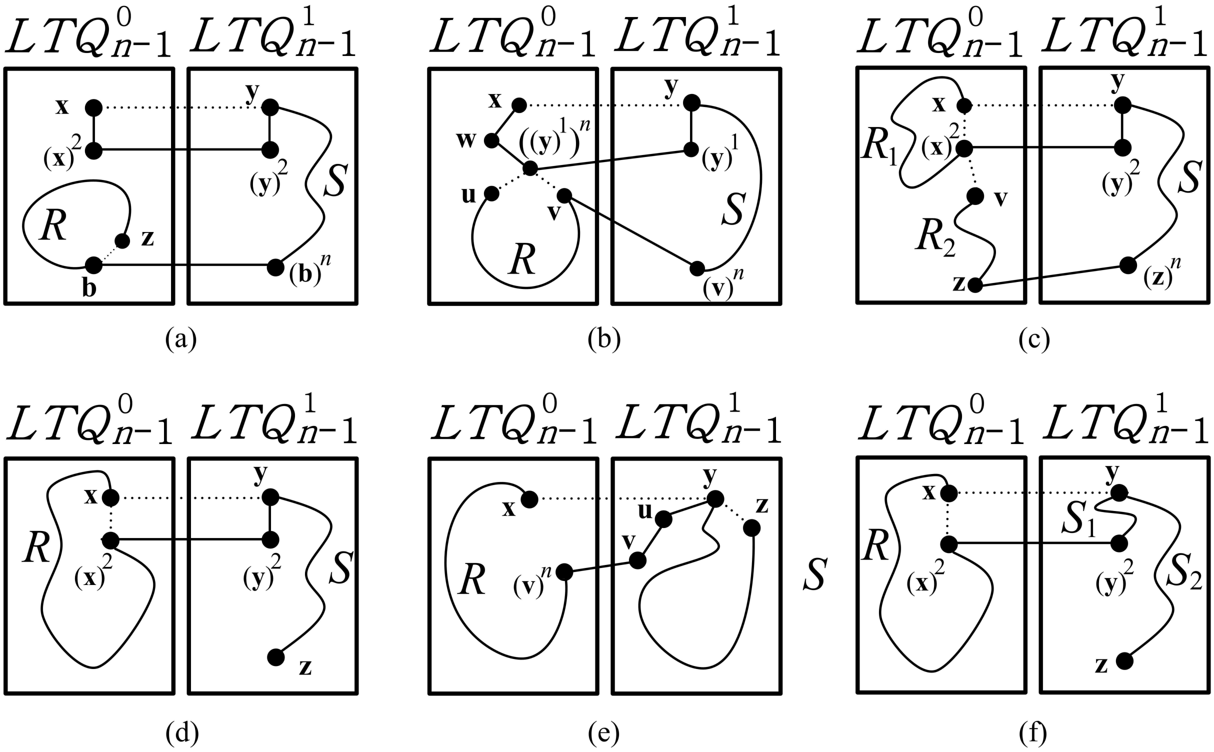

Figure 4: Case 3 in the proof for Theorem 1. (A dashed line or a straight line represents a link.)

4 Conclusion

In this paper, we study the property of hyper-panconnectedness with respect to the class of locally twisted cubes, which receives many researchers attention recently. In a hyper-panconnected network, we can find a path joining any two distinct nodes in a required distance, and this path can be further augmented to form a Hamiltonian path. In terms of such a property, we have more flexibility to design efficient path embedding methods for parallel and distributed computation.

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